Calibration Transfer and Updating in Educational Risk Prediction: A Cross-Cohort Reliability Audit

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ABSTRACT A positive retrospective calibration adaptation gap does not guarantee that updating improves a later cohort. We audit this distinction with OULAD development evidence and complementary XuetangX analyses of an archival indicator and explicitly timed seven-day logged inactivity. Across Logistic Regression (LR), LightGBM and three landmarks, mean Brier adaptation gaps are 0.0166–0.0412 for the archival indicator and 0.0122–0.0261 for baseline inactivity. Full-pipeline recent-active and joint user-assignment sensitivities retain positive gaps, with all 12 adjusted intervals above zero; these overlapping populations are not independent replications. In the archival analysis, a retrospective target-event-rate constant accounts descriptively for 92.4% and 77.0% of Day 7 raw-to-target-OOF improvement for LR and LightGBM. A separate three-presentation test freezes the source predictor, updates calibration using intermediate outcomes, and evaluates later learners unseen in both earlier presentations. All six full-budget full-Platt intervals cross zero after within-population six-comparison adjustment in each of three populations. Exploratory sensitivities identify favorable shrinkage-update signals under selected model, landmark and label-budget conditions, warranting independent validation. Small-budget full updates can also harm future scores. These results support auditing retrospective recoverability and future maintenance utility separately, without selecting a validated update policy. Unverified archival-label timing, restricted risk sets, few course clusters and unknown live log availability limit deployment claims.

INDEX TERMS Calibration transfer, calibration updating, cross-cohort generalization, early academic risk prediction, educational data mining, learning analytics, probability calibration.

I. INTRODUCTION

Early academic risk models use assessment records and digital learning traces to identify learners who may need support. Recent predictive-learning-analytics research spans diverse outcomes, observation windows, and modeling choices [1]. Turning these predictions into timely support also requires an operational process, as illustrated by early-disengagement outreach [2]. Discrimination supports ranking under capacity limits; probability reliability matters when scores are interpreted as risks or used with numerical probability thresholds. A well-ranked learner population can still receive systematically inaccurate risk estimates, and probabilities fitted in one context need not retain their meaning in another.[1]

Transfer across educational settings is already an active research area. Studies have examined cross-institutional performance and fairness [4], [5], course-context differences [6], and the distinction between within-course and within-cohort generalization [7]. Probability calibration is also receiving direct attention: recent OULAD work combines calibration with fairness auditing and selective prediction [8], and Schwerter et al. study calibration alongside transfer across courses and institutions [9]. The practical question is therefore narrower than whether educational models can be calibrated. It is whether a calibration mapping learned from an earlier cohort remains useful when carried forward unchanged.

Calibration transfer has also been studied outside education [10], [11]. A historical cohort can benefit from recalibration while its fitted map transfers poorly; alternatively, the historical map may retain some benefit while a target-adapted map recovers more. These situations have different implications for maintenance. A recent educational preprint uses earlier target-module presentations to recalibrate predictions for a later presentation [12]. That design addresses a useful adaptation setting, but differs from diagnosing the gap between an unchanged historical map and a retrospective map fitted using outcomes from the target population under study.

We explicitly separate source-cohort gain G_S, historical-to-

[1]Drafting assistance: OpenAI Codex [3]; see the Acknowledgment.

future transfer gain G_T, and target out-of-fold gain G_T^*. Their difference $A = G_T^* - G_T$ is a descriptive calibration adaptation gap. It measures additional probability-score utility available to the specified target recalibration procedure beyond the transferred map. It is not a new calibration algorithm or a theoretical bound. Because G_T^* requires target outcomes, it is a retrospective diagnostic rather than a deployable early-warning result.

OULAD supplies development evidence [13]; a separately audited XuetangX distribution supplies repeated-course presentations on a distinct platform [14], [15]. We examine LR and LightGBM with learner-disjoint fitting and calibration roles. The archival label and the explicitly timed inactivity endpoint provide complementary evidence; the latter does not validate the former's unknown generation window. The sensitivity and updating analyses are exploratory, not prespecified confirmation studies.

The contribution has four parts: (1) an empirical audit separating source calibration, unchanged historical transfer, retrospective target OOF adaptation and later-presentation updating; (2) model and landmark comparisons against event-rate and restricted recalibration baselines; (3) outcome and full-pipeline population sensitivities addressing recent activity and repeated users; and (4) future updating analyses across label budgets and conservative strategies, accompanied by course-level uncertainty, influence checks and auditable raw-data execution records. These are evaluation contributions, not a new calibration algorithm or causal identification of distribution shift.

The research questions are:

- **RQ1:** How much Brier-score utility does a historical calibration mapping provide on later same-course presentations, and how does this compare with retrospective target out-of-fold recalibration?
- **RQ2:** How do adaptation gaps vary across model families, prediction landmarks, outcome definitions, and learner risk sets, and how much target-side score improvement is captured by event-rate baselines and intercept-only recalibration?
- **RQ3:** Does recalibration using outcomes from an intermediate presentation improve Brier scores on a later presentation evaluated on learners unseen in both earlier presentations, and how does the observed utility vary with update-label budgets and conservative updating strategies?

II. RELATED WORK AND RESEARCH GAP

A. PREDICTION TIME, OUTCOMES, AND EDUCATIONAL TRANSFER

Prediction tasks in learning analytics differ in what is predicted, when evidence becomes available, and which learners are eligible. Glandorf et al. examine temporal and between-group variability in college dropout prediction [16]. Their results show that predictive performance and predictor relevance vary across observation points and student groups. Pan et al. formulate dropout through survival modeling with explicit historical and prediction windows in a processed MOOC subset [17]. Such work motivates a time-defined endpoint, but its label construction cannot authenticate an unprovided label generator in a different archive. We therefore distinguish OULAD's failure-or-withdrawal outcome and the supplied XuetangX binary label from the time-defined short-term logged-inactivity outcome.[2]

Cross-context prediction can succeed or deteriorate depending on the source-target relationship. Gardner et al. report competitive zero-shot transfer in their four-university setting [4], whereas Yao et al. document performance and fairness losses without localization across a broader university and community-college network [5]. Xu et al. associate performance and fairness shifts with course composition and instructional context [6]. Kleter et al. distinguish generalization across iterations of a course from generalization across courses within a cohort [7]. These studies motivate evaluating a specific deployment relationship rather than assuming universal transfer success or failure. They primarily examine predictor transfer, whereas we focus on the additional mapping applied to its probability outputs.

B. CALIBRATION, SHIFT, AND MEASUREMENT

Post-hoc calibration modifies scores after a base predictor is fitted. Platt's probabilistic mapping [18] and beta calibration [19] are foundational parametric approaches; isotonic regression provides a monotone nonparametric alternative. Our logistic recalibrator acts on the logit of a probability and includes the identity map. It is thus important to distinguish this parameterization from a sigmoid applied directly to arbitrary scores.

Recent shift research explicitly addresses portability. Zhang et al. study transferable temperature calibration for pretrained transformers under noisy text [10]. Roschewitz et al. find that calibration methods targeting shift do not consistently dominate simple methods under the natural image shifts they evaluate [11]. LaSCal estimates calibration without target labels under a specified label-shift assumption [20]. These studies establish that transfer is a methodological concern, while also showing why assumptions, information access, and simple comparison methods must be stated. Our educational setting does not assume that only label prevalence changes.

Multicalibration concerns reliability across multiple subgroups, which is distinct from chronological transfer of a fitted mapping. Hansen et al. show that the benefit of multicalibration post-processing depends on the underlying model's calibration [21]. Wu et al. connect extended multicalibration with out-of-distribution generalization beyond covariate shift [22]. Jin et al. introduce discretization-free multicalibration (DFMC) through tree-ensemble loss minimization [23]. Their guarantees have their own conditions and do not imply that an arbitrary historical calibrator transfers to a later cohort.

[2]Drafting assistance: OpenAI Codex [3]; see the Acknowledgment.

DFMC therefore serves here as a fixed-default robustness comparator.

Calibration error and probability-score utility are also different estimands. Recent distance-to-calibration theory and Smooth ECE address the behavior of calibration measures and binning-based reliability diagrams [24], [25]. Proper-score decompositions separate calibration from discrimination and outcome uncertainty [26]. Our main Brier gain is a held-out score improvement associated with a specified recalibrator; it is not an estimate of every possible calibration defect. Dynamic updating research evaluates another distinct problem: updating models as new outcomes accumulate and testing them on later periods [27]. Target cross-fitting in the present study does not implement such a prospective updating policy.

C. CLOSEST EDUCATIONAL STUDIES AND THE PRESENT CONTRIBUTION

Wu et al. evaluate calibrated OULAD withdrawal models alongside fairness, selective prediction, and capacity-aware alerting under course-presentation group holdouts [8], [28]. Schwerter et al. fit Platt calibration in one course and apply it in other courses within and across two institutions [9]. Li and Tan's recent preprint uses earlier target-module presentations for calibration before evaluation on a later presentation [12]. Their work is still a preprint, and its historical target-data design should not be conflated with our outcome-informed target OOF comparator.

The present contribution extends this evidence to successive same-course presentations by jointly quantifying three utilities under explicit data boundaries. Paired target-side scoring separates the benefit retained by a historical mapping from the additional benefit recoverable after target outcomes become available. Repeated presentations support course-level uncertainty and matched-landmark comparisons; full-pipeline user assignment and a time-defined inactivity endpoint test the pattern on distinct restricted populations. This design contributes an auditable account of where historical information remains useful and where local recalibration adds value. A positive adaptation gap is not evidence that source calibration benefit necessarily decays after transfer: source and target gains concern different populations. We additionally test whether a mapping updated on an intermediate presentation improves a separate later presentation, keeping that maintenance estimand distinct from the retrospective gap.

III. METHODOLOGY AND EXPERIMENTAL DESIGN

A. CROSS-COHORT CALIBRATION TRANSFER FRAMEWORK

Let S denote a historical source course presentation and T a chronologically later target presentation. A base prediction model $f_S(\mathbf{x})$ produces an uncalibrated probability[3]

$$p_i = f_S(\mathbf{x}_i), \qquad p_i \in [0,1]. \tag{1}$$

[3]Drafting assistance: OpenAI Codex [3]; see the Acknowledgment.

A post-hoc calibration mapping $C_S(\cdot)$ is estimated from a source calibration set disjoint from base-model training data. During historical-to-future deployment,

$$\hat{p}_{i,T}^{\text{transfer}} = C_S(p_{i,T}). \tag{2}$$

Learner-disjoint target cross-fitting also produces retrospective target-adapted predictions $C_T^{\text{OOF}}(p_T)$, using fold-specific calibration mappings. Fig. 1 separates this retrospective comparison from a later-presentation updating test.

B. CALIBRATION UTILITY AND ADAPTATION GAP

The primary probability metric is the Brier score:

$$\text{BS} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2. \tag{3}$$

Let L_S be the mean Brier loss on the held-out source calibration pool and L_T the mean Brier loss on the eligible target evaluation set. For domain D, $C_D^{OOF}(p_i)$ denotes the output of the fold-specific calibrator fitted without learner i, rather than one shared fitted map. Positive calibration gain indicates lower Brier loss after calibration.

Source calibration utility is defined as

$$G_S = L_S(p_S) - L_S(C_S^{OOF}(p_S)). \tag{4}$$

Historical calibration transfer utility is

$$G_T = L_T(p_T) - L_T(C_S(p_T)). \tag{5}$$

Target-adapted recalibration utility is

$$G_T^* = L_T(p_T) - L_T(C_T^{OOF}(p_T)). \tag{6}$$

The descriptive calibration adaptation gap is

$$A = G_T^* - G_T. \tag{7}$$

Equivalently,

$$A = L_T(C_S(p_T)) - L_T(C_T^{OOF}(p_T)). \tag{8}$$

A positive A indicates that, with available target outcomes and the specified cross-fitting and calibrator family, target-specific recalibration recovers probability quality beyond the transferred historical mapping. It is neither the entirety of possible calibration need nor a theoretical upper bound. The full source calibration pool fits C_S for G_T; G_S uses source out-of-fold predictions, and G_T^* uses target out-of-fold predictions. Brier gain measures a change in a proper probability score [26], [29]; it does not separately identify the calibration-error component of that score. The gap depends on the chosen calibration family, fitting sample sizes, and fold construction. Source and target gains use different populations, so their magnitudes are descriptive; the two terms defining A are evaluated on the same target learners within each model job.

TABLE 1. Closest calibration-transfer designs and information boundaries. "Not established" means that the cited design does not establish the indicated property; it is not an allegation of leakage.

Study	Calibration data and target-label access	Evaluation relationship and identity boundary	Main contribution
Schwerter et al. [9]	Platt fit on one reference course and applied to another; separate target-prevalence threshold analysis	Cross-course and cross-institution tests; successive same-course cohorts and a cross-course identity refit are not established	Generalization of SRL-aligned prediction and calibration
Li and Tan [12] (preprint)	Earlier target-module presentations supply 10–30% historical calibration samples	Latest presentation reserved for evaluation; learner-disjoint calibration samples reported	Historical target-domain adaptation and capacity-constrained prioritization
Present primary audit	Source-only historical map versus target-label OOF map	Successive same-course pairs; source-present users excluded, with separate full-pipeline cross-course assignment	Joint source/transfer/target utility, prevalence baseline, and heterogeneity
Present three-period check	Source map versus intermediate-period map; third-period labels used only for scoring	Frozen source predictor; third-period learners absent from both earlier presentations	Test whether recent-label updating improves a separate later cohort

C. RECALIBRATION-STRUCTURE DIAGNOSTICS

Three logistic recalibration structures are considered.

Intercept-only recalibration:

$$\operatorname{logit}(p') = \operatorname{logit}(p) + b. \tag{9}$$

Slope-only recalibration:

$$\operatorname{logit}(p') = a \operatorname{logit}(p). \tag{10}$$

Full Platt recalibration:

$$\operatorname{logit}(p') = a \operatorname{logit}(p) + b. \tag{11}$$

The difference

$$G^*_{T,\mathrm{Platt}} - G^*_{T,\mathrm{int}} \tag{12}$$

is used descriptively to measure the incremental gain from allowing both slope and intercept to change instead of the intercept alone. It is not an additive or causal decomposition of the adaptation gap.

D. DATASETS, OUTCOMES, AND TEMPORAL PAIRING

OULAD is the development dataset [13]. We classify Fail or Withdrawn as risk and Pass or Distinction as non-risk. The analysis uses six 2013J-to-2014J course pairs and three 2013B-to-2014B pairs; the four landmarks (Days 7, 14, 28, and 56) and two base models yield 72 pair–landmark–model cells, each repeated over five fixed learner-ID splits. A learner enters a landmark risk set only when registration is dated no later than that day and unregistration is later than that day or absent. Records with missing registration dates, and Withdrawn records lacking an unregistration date, are excluded because eligibility cannot be established reliably. Target learners are excluded if their ID appears in any course module of the full source-season presentation in studentInfo.csv, not only in the corresponding course. The 2014B target also excludes learners retained in the earlier 2013J-to-2014J development analysis. An audit of all 80 fitted season–landmark–model–seed jobs found zero intersections between target IDs and source training, validation, calibration, or their union; all 800 source/target OOF fold checks were also disjoint (Supplementary Section S8). The B season is a retrospective check within OULAD, not a prospectively untouched validation cohort.

XuetangX supplies a distinct-platform retrospective evaluation setting [14]. The analyzed distribution is the Kaggle MOOC Dataset archive [15]; all audit counts refer to this distribution rather than the entire XuetangX platform. The audited archive contains 42,110,402 behavioral events, 225,642 enrollment identifiers, 77,083 unique users, and 247 complete presentation identifiers. Presentations were matched by exact provider and base-course token, ordered by first observed event, and paired with the next observed same-course presentation. Of 84 adjacent candidates, 67 met the prespecified pair-level requirements: at least seven days between the source's last and target's first observed events, at least 300 enrollments per presentation, and at least 50 records of each supplied label class. The final condition uses source and target outcomes, so this is an outcome-dependent analytic sample rather than an outcome-blind deployment population. Further landmark-level support rules, including at least 50 unseen active target learners and at least 10 in the target minority class, yielded 48, 57, and 66 estimable pairs at Days 7, 14, and 28; the full selection flow is in the Supplementary Material. Primary target evaluation retained learners with an event by the landmark who were absent from the immediately preceding source presentation. Requiring an observed event defines an active, behavior-observable risk set, while excluding source-present learners reduces direct learner carryover from the historical calibration presentation. This restriction may limit generalizability to all enrollments; a separate risk-set sensitivity retained all eligible enrollments with zero-filled activity.

The XuetangX archive provides no official course start/end calendar or original label-generation codebook. Each land-

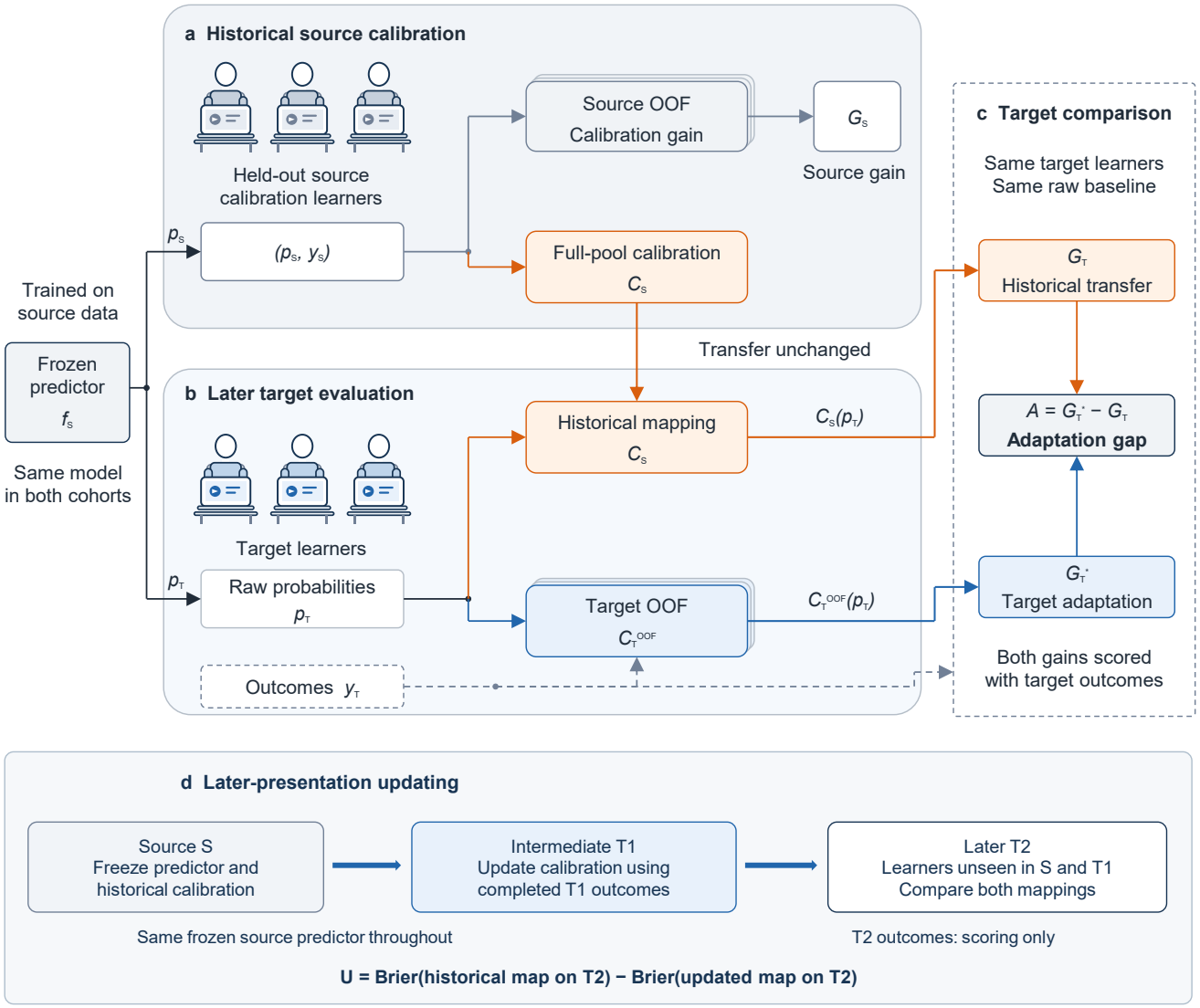


FIGURE 1. Cross-cohort calibration transferability. (a) Held-out source calibration learners support out-of-fold (OOF) estimation of G_S and full-pool fitting of C_S. (b) The same source-trained predictor is frozen in both cohorts. The historical mapping transfers unchanged; target OOF recalibration uses observed target outcomes retrospectively. (c) G_T and G_T^* are Brier gains on the same target learners against the same raw baseline, with $A = G_T^* - G_T$. Target outcomes score both gains but fit neither the frozen predictor nor C_S. Each OOF prediction uses a calibrator fitted without that learner. OULAD pools source calibration observations within season, landmark, model, and seed; XuetangX fits within each source presentation. (d) In a separate three-presentation test, outcomes from T1 update the map while the S predictor stays frozen. Historical and updated maps are evaluated on T2 learners absent from both earlier presentations; T2 outcomes are used only for scoring. Learner symbols are native PowerPoint shapes and do not encode cohort size or demographics.

mark is measured from the first event in that presentation's observed window, not from an official course start. The supplied binary label is analyzed as an archive-provided retrospective risk indicator. Its dropout interpretation is provisional; the negative class is not equated with completion or certification. Without its generation window, we cannot verify that every label refers to an event after the landmark. Although the pair filter required a seven-day observed-activity gap, the minimum analyzed gap was 24.04 days (median 123.02). Neither statistic establishes when the original source labels became available. A separate sensitivity uses an explicit future logged-inactivity endpoint, as described below; it does not replace or authenticate the supplied label. XuetangX therefore provides retrospective cross-presentation evidence, not a verified prospective deployment simulation. These differences in outcome, time origin, and risk set preclude direct comparison of gap magnitudes between platforms. Table 2 summarizes the two protocols.

E. LANDMARK-SPECIFIC FEATURES AND TIME CUTOFFS

All behavioral and assessment features were computed using records timestamped no later than the corresponding prediction landmark. In OULAD, positive-click virtual-learning-environment (VLE) records from Day 0 through the landmark

TABLE 2. Dataset characteristics and historical-to-future evaluation protocols.

Characteristic	OULAD development	XuetangX retrospective evaluation
Outcome	Fail/Withdrawn vs Pass/Distinction	Supplied binary label (window unknown); separate future 7-day inactivity sensitivity
Primary landmarks	7, 14, 28, 56 days	7, 14, 28 days from first observed event
Temporal units	Nine historical–future pairs; 72 pair–landmark–model cells	67 eligible adjacent same-course pairs
Landmark support	Nine pairs at each of Days 7 / 14 / 28 / 56	48 / 57 / 66 pairs at Days 7 / 14 / 28
Selection	Landmark registration and withdrawal eligibility	Pair and landmark support use both label classes
Base models	LR; LightGBM	LR; LightGBM
Calibration	Source 5-fold OOF / full fit; target 5-fold OOF	Source 5-fold OOF / full fit; target 5-fold OOF
Key provenance limit	Institution-specific outcome/protocol	No official course calendar or original truth codebook

provided activity volume, active days, distinct sites, seven-day recency and change, and activity-type click counts. Non-Exam assessments with valid due dates contributed schedule and progress features; only non-banked submissions recorded by the landmark were counted, while assessment scores were never loaded. Pre-course non-banked submissions were separated from in-course volume but could satisfy an already-known due obligation. Published deadlines and weights were treated as known course metadata; missing weights were set to zero. Learners without VLE clicks remained in the risk set with zero activity and an explicit indicator. Course, observable progress-state, and course-by-progress indicators were appended to the 46 numeric OULAD features. Neither outcome labels, withdrawal dates, learner IDs, nor post-landmark records entered the predictor.

XuetangX features used event timestamps from the presentation's first observed event through the landmark, inclusive. They covered event/session/object volume; active-day and within-window temporal engagement; and counts and fractions of action types. For each source split, the action vocabulary was defined only from nonzero counts in source training learners; actions absent there were folded into OTHER in both source and target. Presentation IDs, learner IDs, labels, and future events were excluded from model inputs. Learners with no event by the landmark were absent from primary source fitting and target evaluation; zero-filled all-enrollment analysis was a separate sensitivity. Supplementary Table S1 summarizes the feature families; the reproducibility materials supplies the complete machine-readable inventory.

F. MODELS, CALIBRATION, AND STATISTICAL EVALUATION

The base models were LR and LightGBM [30], with fixed seeds 17, 29, 43, 59, and 71. OULAD source learners were allocated by ID to 60% model training, 20% model validation, and 20% calibration, with roles fixed across landmarks. Its LR candidates had regularization $C = 0.1$ or 1.0; LightGBM candidates had 7 or 15 leaves, with 350 trees, learning rate 0.03, minimum child samples 40, and L2 penalty 5. Each candidate was fitted on the source training set, and the one with lower source-validation Brier score was retained without target-based selection. For LR, count-like features underwent signed log1p transformation and source-training-only standardization. OULAD's source Platt mapping pooled held-out calibration observations across courses within season, landmark, model, and seed; gains were summarized separately by course pair.

For the archival-label analysis, XuetangX roles were 60%/20%/20% by username within each source presentation and fixed across landmarks. For the time-defined outcome, roles were regenerated separately by landmark because the outcome changes with day. Its LR used $C = 1$, L-BFGS, and at most 800 iterations after signed log1p transformation and source-training-only standardization. Its LightGBM used 180 trees, learning rate 0.05, 15 leaves, maximum depth 5, minimum child samples 20, and L2 penalty 1; source validation binary log loss guided early stopping with patience 20. These XuetangX hyperparameters were fixed rather than selected from the two-candidate OULAD grids. Target labels did not tune either base model.

Probabilities were clipped to $[10^{-6}, 1-10^{-6}]$ before logit transformation and scoring. Intercept-only recalibration fixed slope at one and solved for the intercept matching the fitting sample's event rate. Full Platt scaling minimized mean Bernoulli log loss plus a fixed 10^{-5} quadratic penalty around the identity parameters $(a, b) = (1, 0)$, with $a \in [0, 30]$ and $b \in [-30, 30]$. XuetangX slope-only fitting fixed $b = 0$ and minimized the same log loss with $10^{-5}(a-1)^2$. The calibration pool was never used to train or select the base predictor.

To estimate G_S without calibration-set optimism, the source calibration pool was further divided into five learner-disjoint folds. Each learner's calibrated source probability came from a calibrator fitted without that learner. The complete source pool was then used once to fit the transferred C_S for G_T. Five learner-disjoint target folds estimated G_T^* retrospectively. Target outcomes were used to score both target gains, but never to fit the frozen predictor or the transferred source calibrator. OULAD used StratifiedGroupKFold by student ID; XuetangX used stratified folds within a presentation, where each username had one enrollment,

giving learner-disjoint folds. Fig. 1 shows these distinct data boundaries.

Brier gain was primary; log-loss gain, discrimination, calibration-in-the-large, slope, and expected calibration error (ECE) were supporting diagnostics. Calibration-measurement theory cautions against treating arbitrary binning as a definitive calibration test [24], [25]; we did not compute Smooth ECE, and conventional ECE did not define the primary conclusions. We first averaged seeds within each course pair and then weighted pairs equally. Original archival-label XuetangX 95% intervals used 2,000 bootstrap resamples of base-course clusters, accounting for multiple adjacent pairs within a base course. Users may also recur across different base courses; that crossed dependence is not fully represented by this one-way clustering and limits the intervals' inferential interpretation. The fixed common-pair temporal panel was designed after inspecting the primary findings to address changing pair composition; it retains the same pair–seed combinations at all three landmarks and jointly resamples course clusters across time. The later full-pipeline user-assignment sensitivity described below removes observed cross-base-course user overlap before fitting. Its estimates concern a smaller population and do not retrospectively correct the original full-sample intervals. The original primary and associated robustness intervals use pointwise coverage, conditional on the supported pairs and implemented procedure, without multiplicity adjustment. The later updating analysis adds the explicitly defined six- and 96-comparison sensitivities described below. The Supplementary Material reports feature inventories, the outcome-dependent selection flow, absolute target performance, implementation settings, and sensitivity results.

G. FULL-PIPELINE USER ASSIGNMENT AND A TIME-DEFINED OUTCOME

Two exploratory sensitivities address shared-user dependence and uncertain outcome timing. Conditional on the original eligible course pool, each username was assigned to its earliest observed base course, breaking ties lexically without using that user's outcome. This retained 62,142 of 116,363 eligible enrollments; 32,363 of 60,446 eligible users had originally appeared in multiple base courses. The assignment preceded source role allocation, vocabulary construction, preprocessing, base-model fitting, source calibration, target cross-fitting, and evaluation. The original source-target learner exclusion remained in force. This removes observed shared usernames across retained base courses, but does not establish that courses are independent in every other respect.

For the original-label refit, source roles were regenerated using the original 60/20/20 allocation and fixed seeds, while predictor families, settings, and landmark-specific source-role and target support thresholds were retained. Results were compared with the original analysis on common pair-day-model-seed jobs. Matching jobs does not match target learners: original estimates retain their original target membership, whereas refit estimates use the reduced population. Consequently, their difference combines membership, fitting, and calibration changes and cannot identify a pure refitting effect or test equivalence.

The second sensitivity used the same user assignment with a new outcome,

$$Y_{i,d}^{(7)} = \mathbf{1}\{N_i(t_c + d, t_c + d + 7] = 0\}, \tag{13}$$

where t_c is the presentation's first observed event and N_i counts logged events. Features end at $t_c + d$; the outcome records no logged activity in the following seven days. It is short-term logged inactivity, not verified dropout, completion, or permanent disengagement. Because this outcome varies by landmark, source roles were stratified and regenerated separately for each landmark before all models and calibrators were refitted.

This analysis inherits the original 67-pair pool, selected partly using original label-class support, and then applies landmark-specific source-role and target support rules to the new outcome. The pair-level requirements of 300 enrollments and 50 original labels per class are not reapplied after user assignment or relabeling. Observed presentation spans cover the required endpoints; the minimum source outcome-window end to target first-event lead was 46.01, 39.01, and 25.01 days at Days 7, 14, and 28. Observed spans do not certify complete provider logging or actual live label availability. Target OOF calibration remains retrospective. Supplementary Sections S3 and S7 report support, timing audits, and full estimates.

H. COMPLEMENTARY TIME-DEFINED EVIDENCE AND FUTURE UPDATING

These analyses were specified after inspecting the original archival-label results and are exploratory. Throughout, ''primary'' identifies the original analysis protocol, not superior temporal validity of its supplied label. We retain that cohort for retrospective auditing and report the time-defined inactivity stream alongside it, distinguishing each analysis population.

Benefit beyond target prevalence. For each held-out target fold, a constant predictor uses the event rate in the other four folds. It shares the target Platt folds and target-outcome access. Let $\overline{\mathrm{BS}}$ average seeds within pair and then pairs equally. We report

$$\mathrm{BSS}_{null} = 1 - \frac{\overline{\mathrm{BS}}_{Platt}}{\overline{\mathrm{BS}}_{null}}, \qquad \Delta_{ind} = \overline{\mathrm{BS}}_{null} - \overline{\mathrm{BS}}_{Platt}. \tag{14}$$

The ratio is recomputed in every paired cluster resample. This constant uses observed labels from the other target folds. It is a retrospective comparator, not a deployable predictor requiring no target labels. These quantities assess extra score utility beyond that constant; they do not causally attribute improvement to prevalence change.

Course selection independent of the original outcome. We rebuild the candidate pool from all 84 adjacent same-course pairs using only enrollment counts and observed temporal separation, retaining 81 pairs. No supplied truth value or

original-label class threshold enters this selection. We derive the seven-day logged-inactivity outcome for all 247 presentations and refit source roles, preprocessing, predictors, and calibrators. Observed coverage through Day $d + 7$ and source/target class support for the new outcome remain required. This addresses inherited original-label selection, while final estimation still conditions on the new outcome and does not represent all enrollments. Users present in the source presentation are excluded from its target; cross-base-course user assignment is not imposed in this baseline analysis.

Recent-active and joint-assigned populations. We add two full-pipeline sensitivities to the 81-pair structural pool. Recent-active learners must have an event in the preceding seven days, in addition to the baseline activity criterion. The restriction is applied to every source role and later evaluation population before vocabulary construction, preprocessing, predictor fitting and calibration. It coincides with baseline on supported Day 7 jobs. For joint assignment, each user is retained in their earliest observed base course, with lexical tie breaking, before role allocation or fitting. Assignment covers 129 presentations and 48 base courses involved in eligible structural pairs, not the whole platform. It reduces 131,195 eligible enrollments to 65,400, retaining 63,600 users and eliminating observed cross-base overlap within this pool. This changes the study population as well as fitting; matched-job contrasts do not hold learners fixed. The two sensitivities are separate, overlapping restrictions, not independent external replications or their intersection (Supplementary Sections S22–S24).

Three-presentation updating. For each population, the structural pool starts from 33 consecutive sequences $S \to T_1 \to T_2$. A predictor trained only on S scores both later presentations. Historical calibration uses held-out S labels; updated calibration uses a nested random 10%, 20%, 50%, or 100% of eligible T_1 labels. No T_2 label is used in either fit or in budget selection. T1 excludes all source-presentation users. T2 excludes all users present in either earlier presentation, using complete enrollment rosters even for learners absent from fitting subsets. We require each earlier outcome window to end before the next prediction landmark and evaluate

$$U = L_{T_2}\{C_S(f_S)\} - L_{T_2}\{C_{T_1}(f_S)\}. \tag{15}$$

Positive U favors updating. A constant predictor using the same T_1 label subset supplies an additional baseline. This is a temporally ordered retrospective test based on observed timestamps, not a live deployment trial. Budget comparisons on common supported jobs, timing checks, and exclusions are detailed in Supplementary Sections S15–S16.

Conservative and equal-budget updates. The conservative intercept update holds the historical slope fixed and re-estimates only its intercept on T1. A count-weighted alternative shrinks the T1 full-Platt parameters toward the historical parameters,

$$\theta_{shrink} = (1 - w)\theta_S + w\theta_{T_1}, \qquad w = \frac{n_{T_1}}{n_{T_1} + n_{S,cal}}, \tag{16}$$

where n_{T_1} is the realized update-label budget. This fixed rule is a heuristic, not an estimated precision weight. All strategies use identical frozen scores and T2 learners, without T2-based tuning. A separate recency comparison fits full Platt on equally many S and T1 labels, $m = \min(n_{S,cal}, n_{T_1})$, using nested unstratified samples and no replacement of unsupported draws (Supplementary Section S20).

Statistical scope of the sensitivity analyses. Time-defined pair and future-update summaries use 100,000 base-course-cluster resamples (seed 20261012), averaging supported seeds within pair or sequence and then weighting those units equally. A 12-comparison family covers the recent-active and joint-assigned populations' six adaptation gaps each. For updating, percentile tails are expanded to $0.05/(2K)$: $K = 6$ for full-budget full Platt, and $K = 96$ for 72 strategy–model–day–budget contrasts plus 24 equal-budget contrasts. Each updating family is applied separately within each population, not across all archival-label and time-defined analyses. These exploratory Bonferroni-expanded bootstrap intervals have approximate finite-cluster coverage and condition on fitted models, calibrators and selected support. They do not include full-pipeline refitting uncertainty. One million draws check Monte Carlo stability only for four selected tail signals. The original archival-label 2,000-draw results retain their original status. Budget curves use recomputed pointwise intervals on jobs supported at every budget and strategy; actual-support adjusted intervals are reported separately, never transferred onto those curves.

Finite-sample and dependence checks. An equal-budget within-presentation control compares two calibrators trained on disjoint random subsets of the same target presentation and evaluated on a third subset. A source calibrator using the same number of labels is evaluated on that identical subset. Five random repetitions per job separate this random-split reference from the actual temporal contrast. We also inspect every leave-one-base-course-out mean and reweight users and base courses independently, retaining fixed predictions. The latter extends product-weight bootstrap reasoning [31] to a normalized pair-macro sensitivity; it does not propagate training or calibration-fitting uncertainty. Supplementary Sections S17–S19 give definitions, conditional intervals, and calibration plots.

IV. RESULTS

We distinguish two outcome streams. The archival-label results below describe retrospective probability utility for the supplied indicator. The explicitly timed seven-day inactivity results in Sections IV-F–IV-H describe a different behavioral outcome and additionally permit an ordered S–T1–T2 maintenance test. Both streams are central to the audit; their different populations and definitions preclude pooling effect sizes or treating agreement as authentication of the original label.[4]

[4]Drafting assistance: OpenAI Codex [3]; see the Acknowledgment.

A. HISTORICAL CALIBRATION TRANSFER (RQ1)

Calibration support is heterogeneous. Across primary pairs, the median seed-averaged source calibration size is 101.5, 134.4, and 179.6 at Days 7, 14, and 28; the corresponding target medians are 217, 359, and 567.5. Source calibration sizes span 36–600 at Day 7, while targets span 91–2,970. Supplementary Table S14 reports interquartile ranges, positive-label counts, and source/target prevalence. These differences make explicit budget and population checks necessary.

In OULAD, mean source Platt gain was -0.000119 and mean transferred gain was 0.000047 across 72 course–landmark–model units. Supplementary Table S8 separates the J and B seasons, models, and four landmarks. Stratum-level transferred means ranged from -0.00034 to 0.00069; target-adapted gains were generally larger in B than J, and were slightly negative for both J-season models at Day 7. The small overall transfer mean therefore describes a heterogeneous development panel, rather than an absence of course-level variation.

The XuetangX distinct-platform analysis showed base-model dependence. LR retained a positive mean transferred Platt gain at all three landmarks (0.0237 at Day 7 and 0.0036 at Day 28), while LightGBM's mean gain stayed near zero (Fig. 2). LR's transfer-gain intervals were positive at Days 7 and 14, but included zero at Day 28; all three LightGBM intervals included zero (Supplementary Table S9). Positive transfer gain occurred in 45/48, 51/57, and 55/66 LR pairs, compared with 17/48, 29/57, and 30/66 LightGBM pairs. Thus, LightGBM's near-zero average includes both positive and negative course-level gains. At each landmark, target-adapted recalibration recovered greater mean Brier improvement than historical transfer.

B. RETROSPECTIVE ADAPTATION AND RECALIBRATION STRUCTURE (RQ1–RQ2)

OULAD yielded a small mean retrospectively target-adapted Platt gain (0.001971) and adaptation gap (0.001924). In XuetangX, all six primary landmark–model estimates of A were positive, with 95% intervals above zero under the specified base-course-cluster bootstrap (Table 3); the Day 7 LR gap was 0.0412. Absolute Brier scores, target prevalence, and raw AUROC are reported in Supplementary Table S4. At Day 7, raw Brier scores were 0.2080 for LR and 0.1716 for LightGBM; target OOF scores were 0.1432 and 0.1411. Thus, LR's larger calibration gain reflects greater room for improvement and does not establish better final probability quality or deployment suitability. Cross-platform gap magnitudes involve different outcomes and populations.

Retrospectively target-adapted recalibration did not reflect only one form of score adjustment. At XuetangX Day 7, LightGBM's target intercept-only gain was 0.0281 versus 0.0305 for full Platt scaling, whereas LR's was 0.0358 versus 0.0649. The additional full-Platt gain beyond an intercept shift was thus much more prominent for LR, particularly early in the observation window. These differences quantify the incremental value of a two-parameter mapping. Slope-only results and source/target calibration diagnostics are reported in Supplementary Tables S11 and S12, including fixed-slope intercepts, joint-fit slopes, and ten-bin ECE. Secondary log-loss gains gave the same six-condition ordering: target-adapted utility exceeded historical transfer utility in every model–landmark condition (Supplementary Table S2).

C. PROBABILITY QUALITY BEYOND THE TARGET EVENT RATE (RQ2)

Table 4 places the foldwise target-prevalence baseline alongside raw, transferred, and target OOF probabilities. At Day 7, the null reaches Brier 0.1481, compared with 0.1432 for target Platt LR and 0.1411 for LightGBM. Its improvement over raw scores is 92.4% and 77.0% of the corresponding raw-to-target-Platt improvement. These ratios describe aggregate score differences, not a causal fraction explained by prevalence shift. The additional individual-score benefits are 0.0049 (95% interval 0.0026–0.0075) and 0.0070 (0.0044–0.0098), with Brier skill of 0.033 and 0.047. By Day 28 the skills increase to 0.147 and 0.181. Thus, the large early LR correction should not be presented as an equally large gain in individual risk information.

D. LANDMARK AND MODEL HETEROGENEITY (RQ2)

As a exploratory sensitivity addressing changing course composition, we held course pairs and seeds constant across landmarks. The original Day 7, 14, and 28 analyses included 48, 57, and 66 pairs, respectively; their apparent temporal decline could partly reflect this difference. We intersected the estimable pair–seed combinations across all three landmarks and both base models. The resulting matched panel contains 48 course pairs, 37 base-course clusters, and 218 pair–seed combinations, yielding 1,308 pair–landmark–model–seed evaluation instances.

On this fixed panel, LR's A changed from 0.0412 at Day 7 to 0.0344 at Day 14 and 0.0165 at Day 28. LightGBM's values were 0.0307, 0.0280, and 0.0165. The paired Day 7-minus-Day 28 differences were 0.0247 (95% cluster-bootstrap interval 0.0105–0.0405) and 0.0142 (0.0022–0.0291), respectively. Both Day 7-minus-Day 14 intervals crossed zero, while both Day 14-minus-Day 28 intervals were positive. Only about half of individual pairs decreased strictly at both steps; the supported interpretation is a lower late mean in this matched set, not universal monotonic improvement.

LR's target full-Platt gain beyond intercept-only adjustment declined from 0.0290 to 0.0175 and 0.0078 across the matched landmarks; both adjacent paired differences had positive intervals. LightGBM's corresponding incremental gain stayed near 0.002, without a clear temporal contrast. Because the at-risk population, behavior features, and source-fitted base predictor all change with landmark, these contrasts are descriptive rather than causal effects of elapsed course time.

The pair-level gap remained heterogeneous. An exploratory source-observable gate had weak discrimination (AUROC 0.594–0.610) and did not justify a prospective

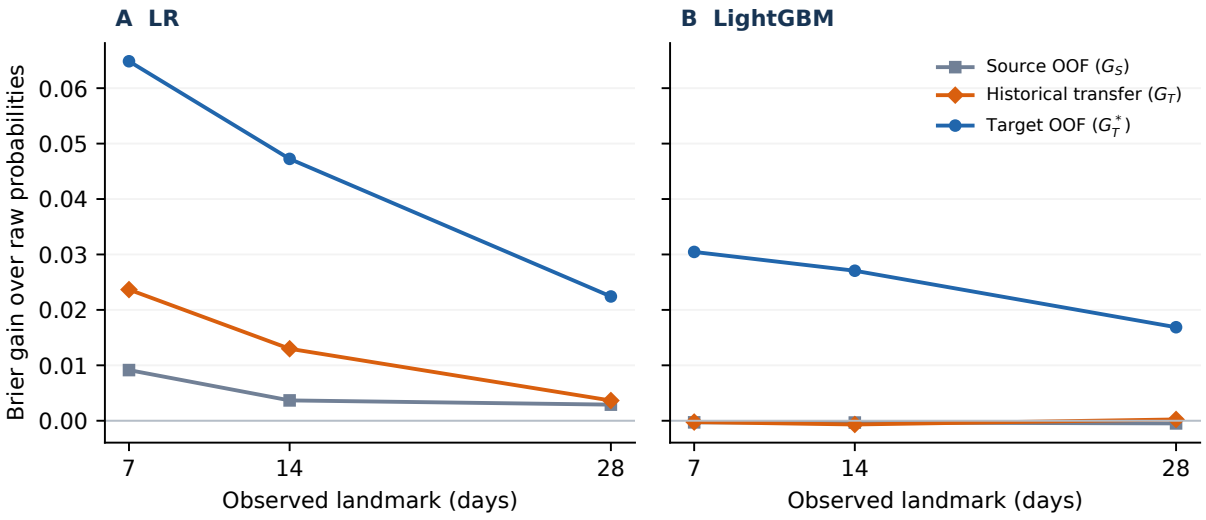


FIGURE 2. Source, transferred, and target-adapted Brier gains across the three primary XuetangX observation landmarks. LR is Logistic Regression. Positive gain means lower Brier loss than raw probabilities; G_T^* requires target outcomes. Pair support varies by landmark; the curves do not identify time effects.

TABLE 3. Primary XuetangX Brier gains. Estimates average seeds within pair and then pairs equally; A intervals resample base courses 2,000 times and do not fully account for users shared across base courses.

Day	Model	Pairs	G_S	G_T	G_T^*	A	95% CI for A
7	LR	48	0.0091	0.0237	0.0649	0.0412	[0.0257, 0.0597]
7	LightGBM	48	-0.0003	-0.0002	0.0305	0.0307	[0.0175, 0.0504]
14	LR	57	0.0037	0.0130	0.0472	0.0343	[0.0198, 0.0518]
14	LightGBM	57	-0.0003	-0.0007	0.0271	0.0277	[0.0139, 0.0442]
28	LR	66	0.0029	0.0036	0.0224	0.0188	[0.0111, 0.0273]
28	LightGBM	66	-0.0005	0.0002	0.0169	0.0166	[0.0098, 0.0248]

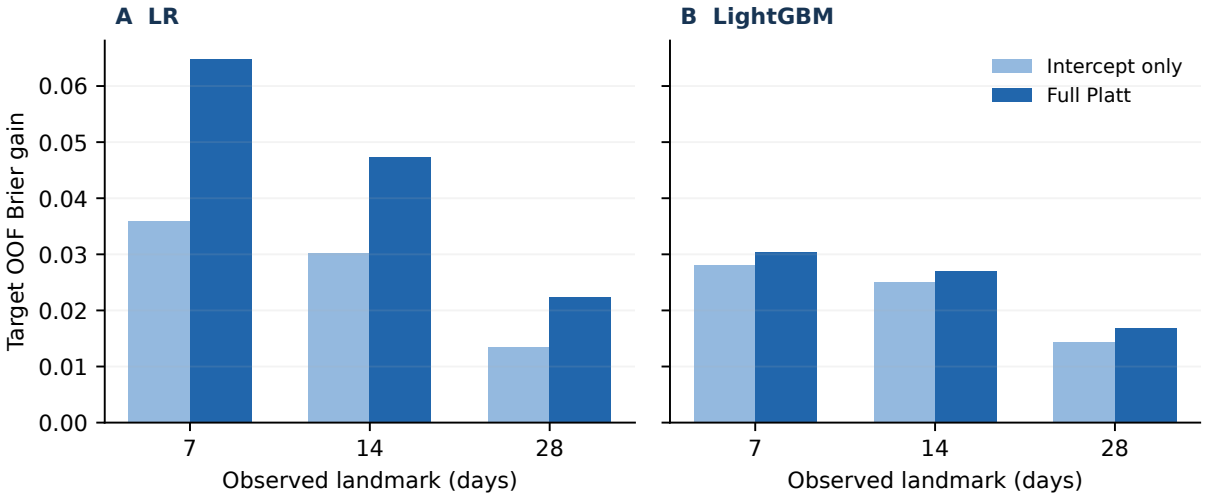


FIGURE 3. Target-adapted intercept-only versus full-Platt Brier gains in XuetangX. Their difference is the incremental full-Platt gain beyond a risk-level shift, not a causal decomposition. LR is Logistic Regression.

calibrator-selection rule; details are in the Supplementary Material.

E. ALTERNATIVE CALIBRATORS AND SUPPORTING CONTROLS (RQ1–RQ2)

Matching the target calibration-label budget to the source calibration size preserved a positive adaptation gap. Beta

TABLE 4. Primary absolute Brier losses and skill relative to the foldwise target-prevalence baseline. Both null and target Platt use target outcomes retrospectively. Skill is a ratio of pair-macro losses, not the mean of pairwise ratios.

Day	Model	Raw	Historical	Null	Target OOF	Δ_{ind}	BSS_{null}	95% CI: BSS
7	LR	0.2080	0.1844	0.1481	0.1432	0.0049	0.033	[0.018, 0.049]
7	LightGBM	0.1716	0.1718	0.1481	0.1411	0.0070	0.047	[0.031, 0.066]
14	LR	0.1814	0.1685	0.1450	0.1342	0.0108	0.074	[0.054, 0.097]
14	LightGBM	0.1574	0.1581	0.1450	0.1303	0.0146	0.101	[0.078, 0.125]
28	LR	0.1488	0.1451	0.1481	0.1264	0.0217	0.147	[0.119, 0.178]
28	LightGBM	0.1381	0.1379	0.1481	0.1213	0.0268	0.181	[0.148, 0.217]

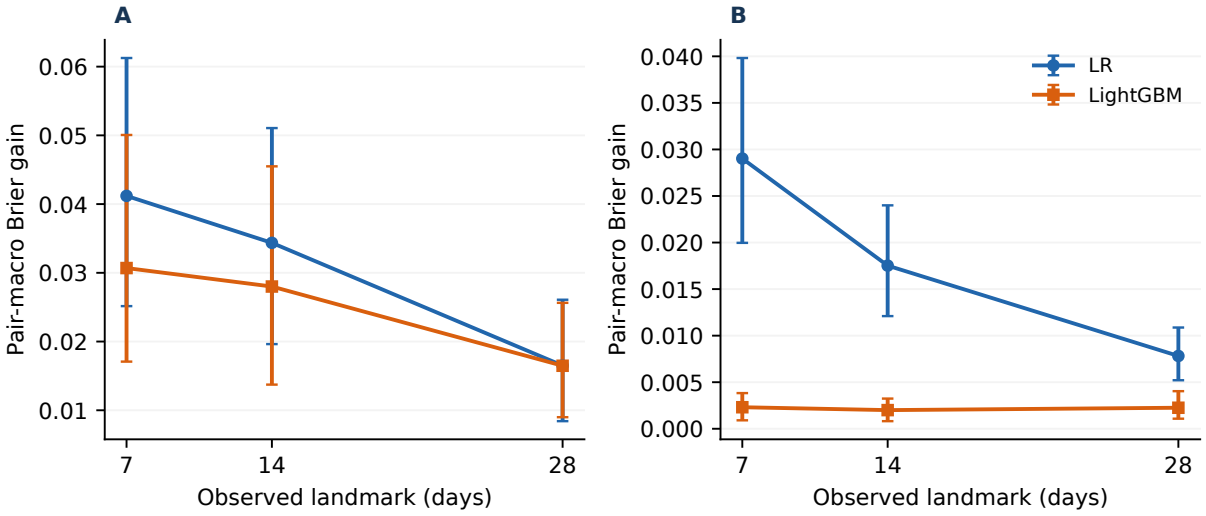


FIGURE 4. Exploratory matched-course sensitivity on 48 common pairs, 37 base-course clusters, and 218 pair–seed combinations at Days 7, 14, and 28. Blue circles denote LR and orange squares LightGBM. Panel A plots $A = G_T^* - G_T$; panel B plots target OOF full-Platt minus intercept-only gain. Error bars are 95% percentile intervals from base-course-cluster bootstrap resampling.

TABLE 5. Matched-course temporal sensitivity. Identical pair–seed combinations are retained at all three landmarks.

Day	Model	G_T	G_T^*	A	95% CI for A	$G^*_{T,\mathrm{full}} - G^*_{T,\mathrm{int}}$
7	LR	0.0237	0.0649	0.0412	[0.0252, 0.0613]	0.0290
7	LightGBM	-0.0002	0.0305	0.0307	[0.0171, 0.0501]	0.0023
14	LR	0.0162	0.0506	0.0344	[0.0196, 0.0511]	0.0175
14	LightGBM	0.0013	0.0294	0.0280	[0.0137, 0.0455]	0.0020
28	LR	0.0051	0.0216	0.0165	[0.0084, 0.0261]	0.0078
28	LightGBM	-0.0001	0.0164	0.0165	[0.0090, 0.0256]	0.0023

calibration, isotonic regression, and fixed-default DFMC also showed target-adapted utility exceeding historical transfer utility (Supplementary Table S6). Retraining on an all-enrollment zero-filled risk set reduced some early gaps but did not eliminate them. Earliest-to-latest same-course pairing yielded the same direction among the subset of courses with at least three eligible offerings; absolute values across these different populations should not be compared directly.

Same-course historical mappings had higher mean transfer gain than the random earlier unrelated-course control in all six conditions (Supplementary Table S10). Paired difference intervals were positive for LR at Day 7 and LightGBM at all three landmarks; LR at Days 14 and 28 remained inconclusive. The contrast evaluates the implemented historical-source choice. Because the calibrators were learned from separately trained base models with potentially different score distributions, it does not isolate an intrinsic course effect.

The within-presentation random-split gap is close to zero in every primary condition (means -0.0010 to 0.0001, all intervals spanning zero). Matched-budget temporal gaps remain larger, with temporal-minus-random contrasts from 0.0148 to 0.0417 and all pointwise intervals positive. This weakens a pure finite-sample-estimation explanation under the specified control. It does not identify which aspect of the source–target relationship changed (Supplementary Table S17).

All leave-one-base-course-out adaptation-gap means re-

main positive. Conditional user–course reweighting also retains positive intervals in all six conditions (Supplementary Table S18). These checks do not resolve repeated users' influence on shared fitted models or establish independent-course sampling. Reliability plots selected by target sample size, rather than favorable gain, and post-calibration intercept/slope diagnostics show residual imperfections: at Day 7, median target OOF joint-fit slopes are 0.633 for LR and 0.753 for LightGBM, below one. Better Brier scores therefore do not imply perfect calibration (Supplementary Table S19 and Fig. S1).

F. OUTCOME AND FULL-PIPELINE POPULATION SENSITIVITIES (RQ2)

The archival-label user-assignment refit retains positive gaps on 908 jobs and 19/36/44 pairs at Days 7/14/28. Its matched-job difference intervals include zero, which establishes neither equivalence nor a fitting-only effect. The user-assigned inactivity sensitivity uses 1,220 jobs but inherits the original 67-pair selection (Supplementary Sections S3 and S7). The supplied label agrees with future seven-day inactivity on 72.7%, 76.3% and 82.5% of active enrollments by landmark, but this does not establish its generation rule or timing.

Rebuilding from the structural pool yields 73/79/80 supported inactivity pairs and 2,186 baseline model fits. Mean A spans 0.0122–0.0261. The recent-active full refit retains 73/75/67 pairs and 1,932 fits; its gaps span 0.0163–0.0280. Joint assignment retains 34/53/56 pairs and 1,256 fits; its gaps span 0.0125–0.0314. All 12 population-sensitivity A intervals remain above zero after their joint 12-comparison expansion (Table 6). These findings retain an adaptation gap under restricted risk sets and reduced observed user dependence; they do not identify a causal shift mechanism. Recent-active Day 7 duplicates baseline supported jobs and supplies no independent replication. Joint assignment is conditional on the eligible course pool, and neither restriction authenticates the archival outcome or guarantees complete logging.

G. UPDATING UTILITY IN A LATER PRESENTATION (RQ3)

Baseline updating supports 30/32/32 sequences across 23/24/24 base-course clusters, with 896 source-model fits. Earlier outcome windows end at least 56.07 days before the next prediction landmark. Table 7 reports absolute T2 performance, including raw and historical scores. All strategies share the source predictor and evaluation learners; T2 labels never fit or select an update. The six full-budget full-Platt means range from -0.0010 to 0.0135. Only Day 14 LR has a positive pointwise interval, 0.0135 [0.0005, 0.0285]; its six-comparison interval is [-0.0033, 0.0345]. All six adjusted intervals cross zero.

Recent-active and joint-assigned rollouts require 828 and 512 additional fits. Their full-budget six-comparison intervals also all cross zero, separately within each population (Supplementary Tables S25–S26). Joint assignment leaves only 10/23/26 sequences and 8/19/21 clusters at Days 7/14/28, so especially the Day 7 intervals are weak evidence about wider populations. The 7,610 total pair and rollout fits across all three populations are repeated computations, not independent cohorts.

Small update sets can harm future scores. In the baseline Day 7 actual-support analysis, the 10% budget uses a median 21 labels and produces mean $U = -0.0161$ for LR and -0.0237 for LightGBM. These negative pointwise intervals do not remain below zero under the baseline 96-comparison expansion. Unsupported samples are retained as unsupported without redrawing. Figure 5 uses a fixed common-support panel to compare budget trajectories. Full-budget individual updates outperform the T1 event-rate constant in all six baseline conditions (Supplementary Section S16), but this does not establish superiority to the unchanged historical map.

H. CONSERVATIVE UPDATING AND COURSE HETEROGENEITY (RQ3)

In the baseline analysis, intercept updating and count-weighted shrinkage reduce some early small-budget losses; shrinkage has the largest full-budget mean in all six conditions. Nevertheless, all baseline 96-comparison intervals include zero, including the equal-label-budget S/T1 comparisons. Ranking point estimates does not establish a preferred updating policy or a causal benefit of recency.

The recent-active and joint-assigned populations reveal four exploratory tail signals under their own 96-comparison families. Recent-active LR shrinkage is favorable at Day 14 with 20% labels, $U = 0.0136$ [0.0009, 0.0406] (24 clusters), and at Day 28 with 10%, 0.0064 [0.0004, 0.0209] (20 clusters), or 20%, 0.0071 [0.0003, 0.0182] (23 clusters). Joint-assigned Day 14 LightGBM full updating is unfavorable at 10%, -0.0353 [-0.0995, -0.0023] (14 clusters). One-million-draw tail checks retain these directions; the reported intervals remain the 100,000-draw estimates. The positive lower bounds are close to zero, the course counts are modest, and correction is population specific. These selected signals warrant independent validation, not a claim of statistically established general shrinkage benefit (Supplementary Table S27).

Course-level utility is heterogeneous. For baseline full-budget Day 14 LR, 62.5% of base-course means are positive and course means range from -0.0355 to 0.1486. Deleting one course gives overall means between 0.0091 and 0.0168, while the six-comparison interval still crosses zero. Conversely, baseline Day 7 and Day 28 LightGBM overall means can change sign after a single-course omission. We report every omission rather than selecting a favorable subset (Supplementary Table S28 and Fig. S3). Positive retrospective A is therefore insufficient to guarantee later $U > 0$; we have not formally tested or excluded a statistical predictive relationship between A and U.

V. DISCUSSION

TABLE 6. Additional time-defined outcome populations (RQ2). P/C denotes supported pairs/base-course clusters; A intervals use 100,000 cluster resamples and a single 12-comparison family spanning the two restricted populations, three days and two models. Recent-active Day 7 repeats the baseline on supported jobs and is not independent confirmation. Joint assignment covers the eligible structural course pool only.

Day	Model	Recent P/C	Recent-active *A* [adjusted interval]	Joint P/C	Joint-assigned *A* [adjusted interval]
7	LR	73/47	0.0261 [0.0159, 0.0400]	34/28	0.0238 [0.0139, 0.0368]
7	LightGBM	73/47	0.0174 [0.0096, 0.0286]	34/28	0.0174 [0.0086, 0.0273]
14	LR	75/47	0.0280 [0.0181, 0.0405]	53/41	0.0314 [0.0169, 0.0521]
14	LightGBM	75/47	0.0228 [0.0138, 0.0341]	53/41	0.0171 [0.0100, 0.0260]
28	LR	67/43	0.0216 [0.0112, 0.0346]	56/42	0.0175 [0.0097, 0.0272]
28	LightGBM	67/43	0.0163 [0.0091, 0.0265]	56/42	0.0125 [0.0069, 0.0196]

TABLE 7. Baseline three-presentation absolute T2 performance (RQ3), using the full T1 label budget. Values average seeds within sequence and sequences equally. N/C denotes sequences/base-course clusters. T2 event rate is descriptive and never used for updating. All six full-Platt U intervals cross zero under the six-comparison sensitivity (100,000 resamples); Supplementary Tables S25–S26 report all populations and uncertainty.

Day	Model	N/C	Event rate	Raw BS	Historical BS	Full BS	Intercept BS	Shrink BS
7	LR	30/23	0.6416	0.2435	0.2238	0.2182	0.2229	0.2162
7	LightGBM	30/23	0.6416	0.2129	0.2108	0.2095	0.2111	0.2063
14	LR	32/24	0.7338	0.2003	0.1878	0.1744	0.1757	0.1715
14	LightGBM	32/24	0.7338	0.1712	0.1726	0.1666	0.1682	0.1638
28	LR	32/24	0.7797	0.1569	0.1485	0.1447	0.1442	0.1414
28	LightGBM	32/24	0.7797	0.1334	0.1345	0.1355	0.1359	0.1333

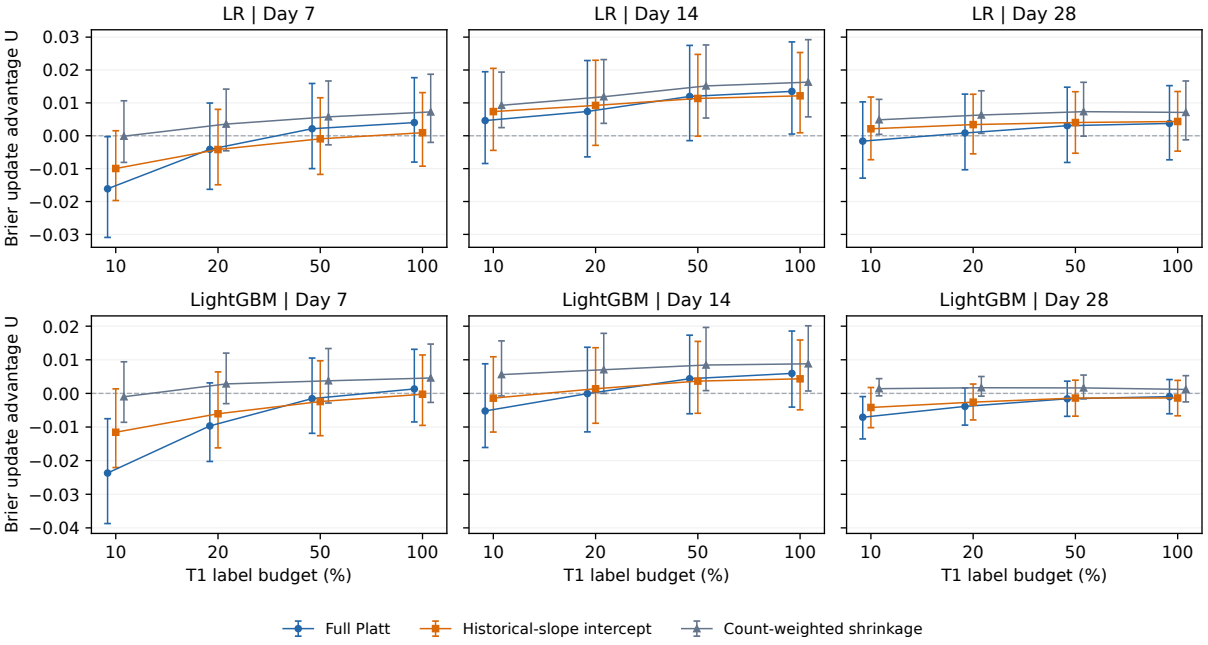


FIGURE 5. Baseline third-presentation Brier advantage over the unchanged full S map. All budgets and strategies use identical supported model jobs and frozen source predictors. Error bars are recomputed 100,000-draw pointwise course-cluster intervals; no actual-support multiplicity marker is imported. Day 7 includes 29 sequences and 22 clusters; Days 14 and 28 each include 32 sequences and 24 clusters. This differs from the Day 7 full-budget support in Table 7.

A. HISTORICAL CALIBRATION IS INFORMATIVE BUT INCOMPLETELY TRANSFERABLE

Historical calibration transfer utility differed markedly by underlying model. OULAD showed little mean transferred Platt gain, whereas XuetangX LR retained positive transfer gain and LightGBM generally did not. Target-adapted recalibration still exceeded historical transfer on average in the six primary XuetangX conditions. The target-side gap concerns the adequacy of a transferred relationship. For example, LR's Day 7 transferred gain (0.0237) exceeds its source OOF gain (0.0091), despite remaining below its target OOF gain (0.0649). The observed gap therefore does not establish universal source-to-target decay or blanket failure of historical

recalibration.[5]

The same-course historical control showed positive mean differences from random earlier unrelated-course mappings, with inconclusive LR contrasts at Days 14 and 28. Relevant historical data can contain useful calibration information even when the previous mapping is incomplete. The control does not isolate a course-specific law: unrelated courses also had separately fitted base predictors and different score distributions. Relevance of the historical source and sufficiency of its calibrator are therefore distinct questions.

The evidence motivates treating calibration utility as a property of the model–cohort deployment relationship, rather than a fixed attribute of the fitted model alone. This view fits broader concerns about calibration under shift [11], [22], [32] and the dependence of post-processing benefit on the underlying predictor [21]. Proper scoring rules assess the quality of probability forecasts rather than only their ranking [29]. In early-warning applications, changing cohort composition and outcome prevalence can alter the decision meaning of a nominal risk threshold; operational studies also show why prediction outputs need to be connected to actual support workflows [2], [8].

B. RETROSPECTIVE UTILITY AND FUTURE UPDATING ARE DISTINCT

The intercept/full-Platt comparison suggests different forms of retrospectively recoverable utility across base models. In XuetangX, an intercept shift captured much of LightGBM's target-adapted gain. LR gained more from jointly re-estimating intercept and slope, especially at early observed landmarks. This is an incremental fit comparison under one calibrator family, not identification of a unique data-generating mechanism.

Model complexity alone cannot explain the difference. LR and LightGBM produce different raw score distributions, and their training objectives, regularization, and response to sparse early behavior also differ. The intercept and slope estimates summarize how those scores relate to observed outcomes in the target cohort; they should not be interpreted as causal effects of model family or as a guarantee for every future presentation. Calibration assessment likewise distinguishes reliability from discrimination and outcome uncertainty [26], [32].

The original archival-label matched-pair panel supports a lower late retrospective gap on fixed course pairs, while the three-presentation inactivity experiment addresses a different maintenance question. Conservative updates lessen some early small-budget losses, and selected recent-active LR conditions show favorable exploratory signals. The positive adjusted lower bounds are close to zero and arise within population-specific families; they do not establish that lower parameter flexibility or greater recency guarantees future benefit. The count-weighted mixture is one untuned heuristic; these data do not identify an optimal shrinkage rule, a causal drift mechanism, or a threshold for safe updating.

C. IMPLICATIONS, LIMITATIONS, AND FUTURE DIRECTIONS

Probability calibration should be audited as new cohorts produce outcomes. Raw and historically calibrated scores should both be retained so Brier score, log loss, calibration-in-the-large, and calibration slope can be reassessed. The target-adapted quantity G_T^* requires target labels and is a retrospective diagnostic under the specified cross-fitting protocol. It is not a prospective gain, a sequential updating policy, or an attainable upper bound. Weak exploratory gate discrimination gives no basis for automatic source-only calibrator selection. We did not evaluate intervention benefit, resource allocation, or effects on learner outcomes. Improvement under one proper score does not guarantee utility for every downstream decision maker [33].

Several provenance and selection boundaries remain. XuetangX landmarks begin at first observed activity, not an independently verified course start. The original truth-generation rule and timing remain unavailable. The seven-day logged-inactivity outcome supplies an explicit future boundary without certifying the original label or complete event collection. The 1,220-job sensitivity inherits original-label selection; the structural-pool baseline and population refits remove that pair-selection dependency. All still require support for the time-defined outcome. Recent-active selection can emphasize a different stage of engagement, while joint assignment discards later cross-course participation; neither is representative of every platform learner. Pair and landmark eligibility partly depend on target class counts; results describe supported repeated-course pairs rather than all 247 complete presentation identifiers. OULAD and XuetangX differ in outcomes, anchors, populations, and risk sets, preventing direct interpretation of their gap magnitudes as platform effects.

Full-pipeline user assignment removes observed shared-user overlap across retained base courses and preserves positive adaptation gaps, but substantially changes the study population. Only 19 original-label pairs remained at Day 7, and the joint-assigned Day 7 updating analysis has eight course clusters. Assignment applies within each defined eligible pool, not platform-wide. Original/refit contrasts match jobs rather than identical target learners, and intervals crossing zero are not equivalence evidence. The restricted refit does not retrospectively validate the original full-sample intervals, whose one-way clustering omits crossed user dependence. Multiple pairs within a base course are clustered, but shared platform conditions and other course dependence may remain. Seeds and cross-fitting stabilize descriptive estimates; they do not create new independent cohorts. Pointwise intervals and multiple exploratory sensitivities should be interpreted as supporting analyses.

The three-presentation test shows why retrospective adaptation cannot justify automatic updating. Even with completed earlier outcome windows, recent calibration labels do

[5]Drafting assistance: OpenAI Codex [3]; see the Acknowledgment.

not consistently improve the next cohort, and small budgets can worsen scores. Its original pointwise intervals are complemented by six- and 96-comparison sensitivities, but baseline inference rests on 23–24 clusters and joint-assigned inference on only 8–21. The latter removes observed cross-base user overlap within the eligible pool, but shared platform conditions and archive-arrival uncertainty remain. All cluster intervals are conditional on already fitted models and omit refitting uncertainty. Multiplicity adjustment widens uncertainty; it cannot turn these exploratory tests into confirmatory evidence or establish absence of an effect. A future prospective study should verify live data delivery and evaluate a prespecified update-or-retain policy on further cohorts. Dynamic model-updating research provides relevant designs [27]. The fixed count-weighted shrinkage examined here motivates evaluating historically selected hierarchical pooling on additional cohorts, when score scales are comparable; no pooling hyperparameter should be chosen from the eventual test cohort. Label-shift methods such as LaSCal offer another comparator when their assumptions and target-data requirements are appropriate [20]; our study does not establish that educational shift is label shift. Prospective evaluation should also test whether source-observable indicators predict calibration deterioration beyond the weak gate reported here and whether better probabilities improve support decisions.

VI. CONCLUSION

Retrospective adaptation gaps persist across two complementary XuetangX outcomes and several restricted populations, including recent-active and jointly assigned learners. These gaps describe additional score utility available to target OOF recalibration; event-rate and intercept-only baselines clarify how much of that improvement requires individual score adjustment, without identifying a causal drift mechanism. Positive retrospective gaps do not guarantee that intermediate-label updating improves a later cohort: full-budget full-Platt intervals all cross zero after population-specific six-comparison adjustment. Selected recent-active LR shrinkage conditions offer exploratory potential, while a small-budget joint-assigned LightGBM condition shows harm. Neither pattern establishes a validated deployment policy. Future maintenance should compare updating with retention on further learners unseen during fitting, under prespecified strategies and verified data availability. Original-label timing, restricted support, limited clusters and conditional inference remain substantive boundaries.[6]

DATA AND CODE AVAILABILITY

OULAD is distributed by the Open University [13]; the analyzed XuetangX archive is the Kaggle distribution [15]. Analysis code, protocols and dependency specifications are available at https://github.com/13406330468/Educational-Risk-Prediction. The analysis-source snapshot is identified by commit `538ebd0`. Frozen aggregates and the numeric evidence registry are distributed separately in `Frozen_Results.zip`; Section S13 specifies its checksum and extraction step. Raw records, learner identifiers and individual predictions are excluded. Repeated execution from the original XuetangX logs covered the key two- and three-presentation time-defined pipelines, totaling 7,610 base-model fits, with support and numerical comparisons. This is a separate execution of the same algorithms, not a different implementation or independent-team replication; it does not reproduce all OULAD or other archival-label analyses. Aggregate regeneration is separately identified.

ETHICS AND DATA GOVERNANCE

This study is a secondary analysis of publicly available educational datasets. It involved no recruitment, intervention, direct contact with learners, or collection of new personal data. The authors considered additional ethics review and consent unnecessary for this secondary use of publicly released records; no institutional approval or formal exemption is claimed. Supplied pseudonymous identifiers were used only to enforce data separation and audit repeated users. No attempt was made to identify individuals. The public reproducibility package contains code and aggregate results, with learner-level records and predictions excluded.

CONFLICT OF INTEREST

The authors declare no financial or nonfinancial conflicts of interest related to this work.

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